

SIMATS ENGINEERING

ASSESSMENT TOOL 3

NAME: Aslesha Surya Kiran M

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COURSE NAME: CSA1603

COURSE CODE: Data Warehousing and Data Mining

COURSE OUTCOME COVERED

CO4: Examine the applicability of clustering algorithms in real-world scenarios, presenting findings through clear reports with effective visualizations.

Assessment Tool Weightage: 20%

QUESTIONS: 05

Total Marks:50

Annexure A

COMPARITIVE ANALYSIS

<i>Criteria</i>	Scope of Items (2)	Comparison Depth(2.5)	Use of Evidence (2)	Critical Thinking (2)	Clarity of Presentation (1)	<i>Total(10)</i>
<i>Weightage</i>	20%	25%	20%	20%	10%	<i>100%</i>
<i>Q1</i>						
<i>Q2</i>						
<i>Q3</i>						
<i>Q4</i>						
<i>Q5</i>						

Code of Conduct:

I _M A Surya Kiran/192411236_ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

m A SURYA kirAn

Signature of the student:

RUBRICS FOR CALCULATION:

		Comparative analysis			
Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Scope of Items	20%	Selects highly relevant items with clear scope and justification	Selects appropriate items with minor gaps	Selection is limited or partially relevant	Items are unclear or not relevant
Comparison Depth	25%	Provides detailed and comprehensive comparison across multiple aspects	Provides adequate comparison with some depth	Limited comparison with few aspects	Very minimal or no comparison
Use of Evidence	20%	Supports comparison with accurate data, examples, or references	Uses some supporting evidence with minor gaps	Limited or weak supporting evidence	No supporting evidence provided
Critical Thinking	20%	Demonstrates strong critical thinking with clear interpretation and reasoning	Shows reasonable interpretation with minor gaps	Basic interpretation; lacks depth	No meaningful interpretation
Clarity of Presentation	15%	Well-organized, clear, and logically structured presentation	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand

Annexure A

Q1 . Dataset: Assessment Tool 3: Comparative Analysis (25% → 5 Marks)

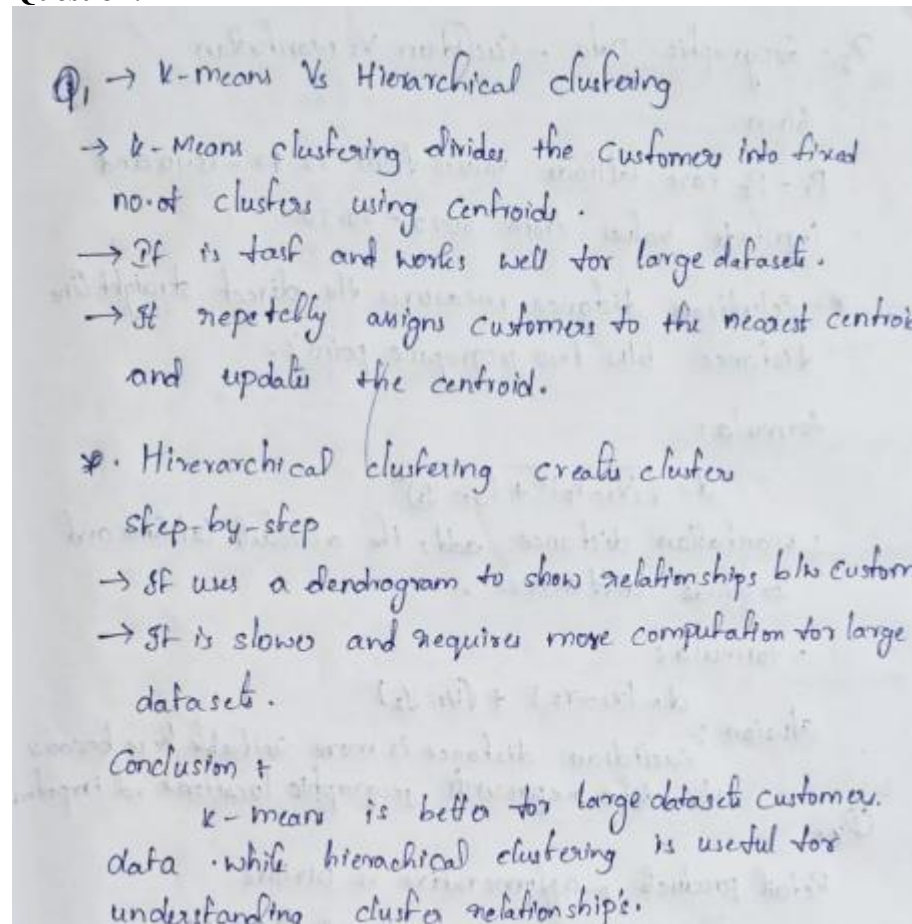
Time duration :45 Minutes
Assessment Tool 3 (Low)

Q1 . Dataset: Customer Data

Customer Income (₹) Spending Score

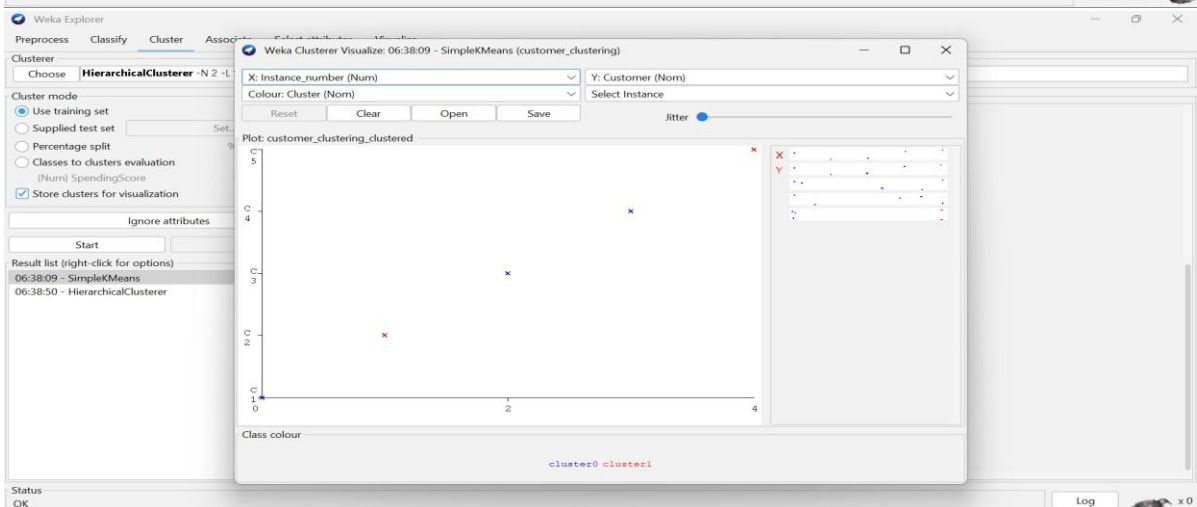
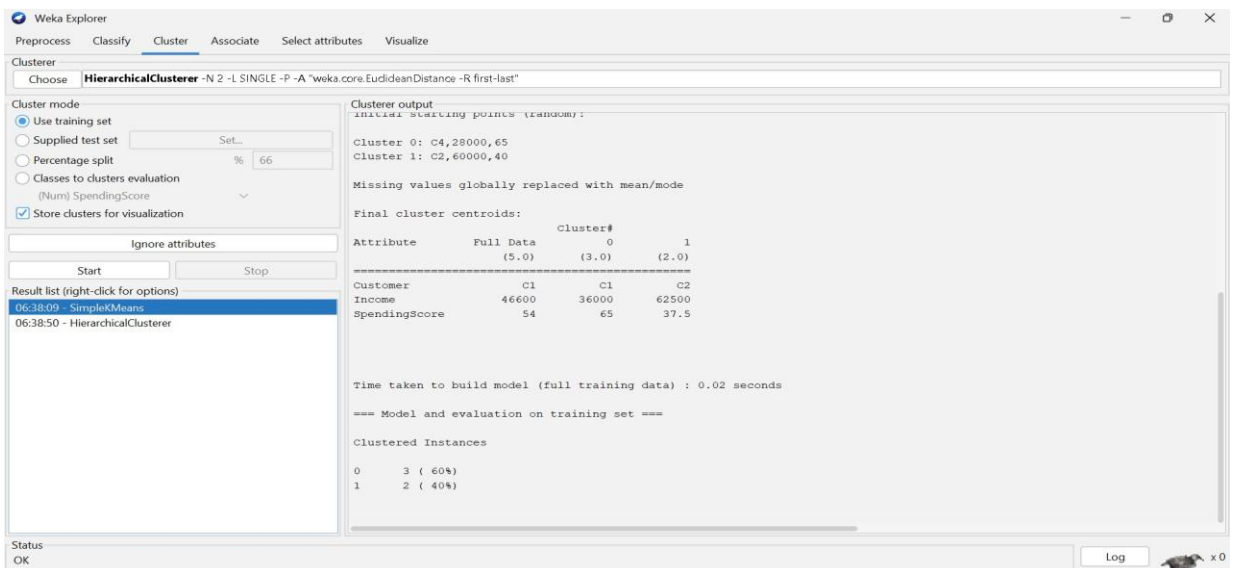
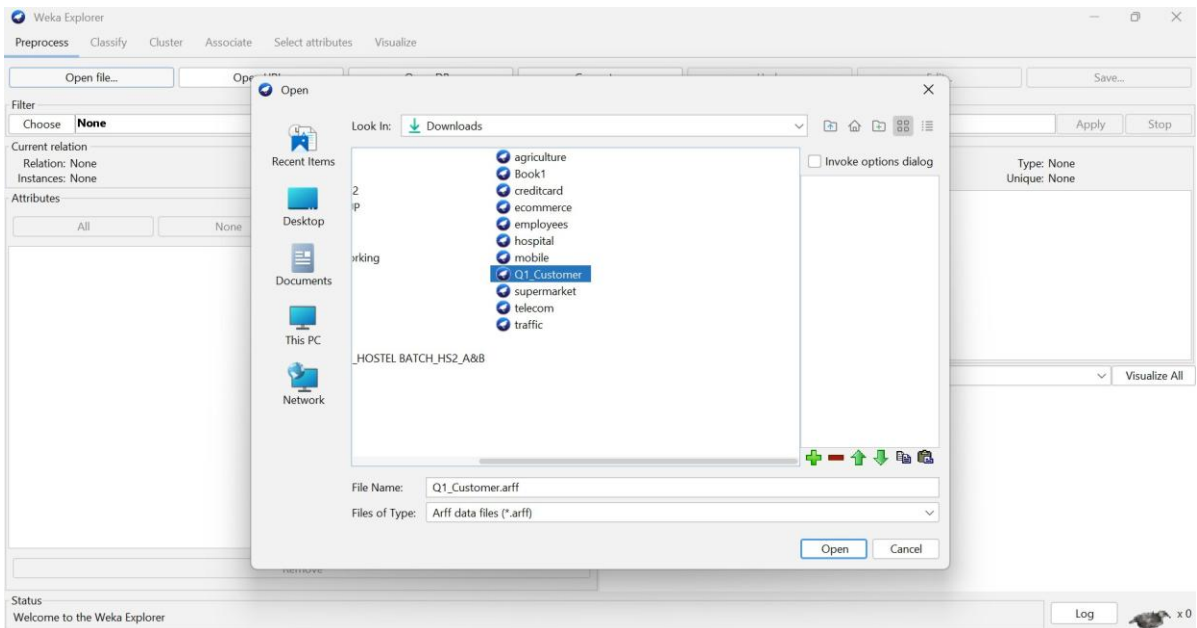
C1	30000	60
C2	60000	40
C3	50000	70
C4	28000	65
C5	65000	35

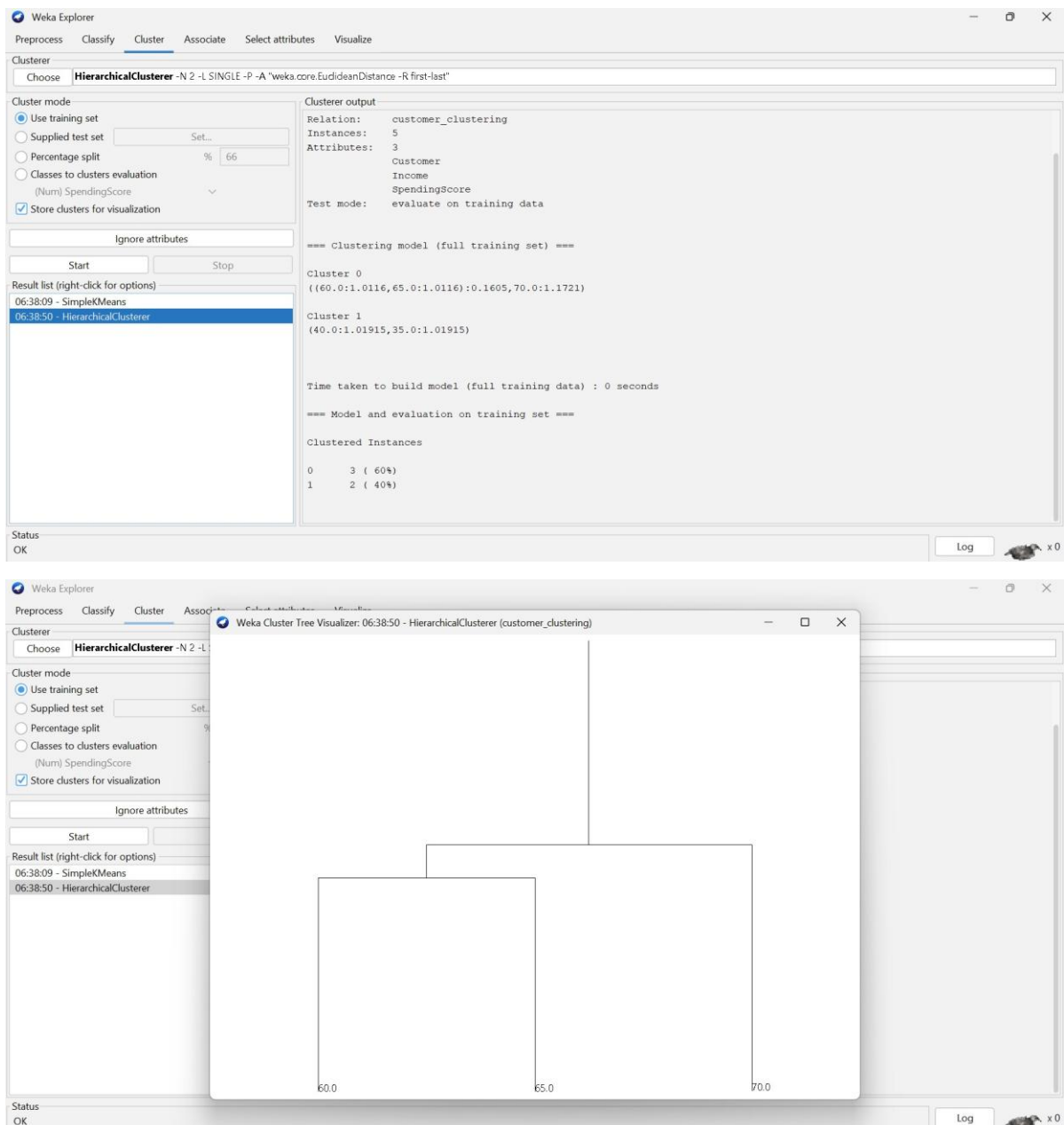
Question:



Compare K-means and hierarchical clustering using this dataset and explain differences in performance and scalability.

OUTPUT:





Q2. Dataset: Document Data

Document Word1 Word2 Word3

D1	0.2	0.5	0.3
D2	0.1	0.6	0.3
D3	0.8	0.1	0.1
D4	0.75	0.15	0.1
D5	0.2	0.4	0.4

Question:

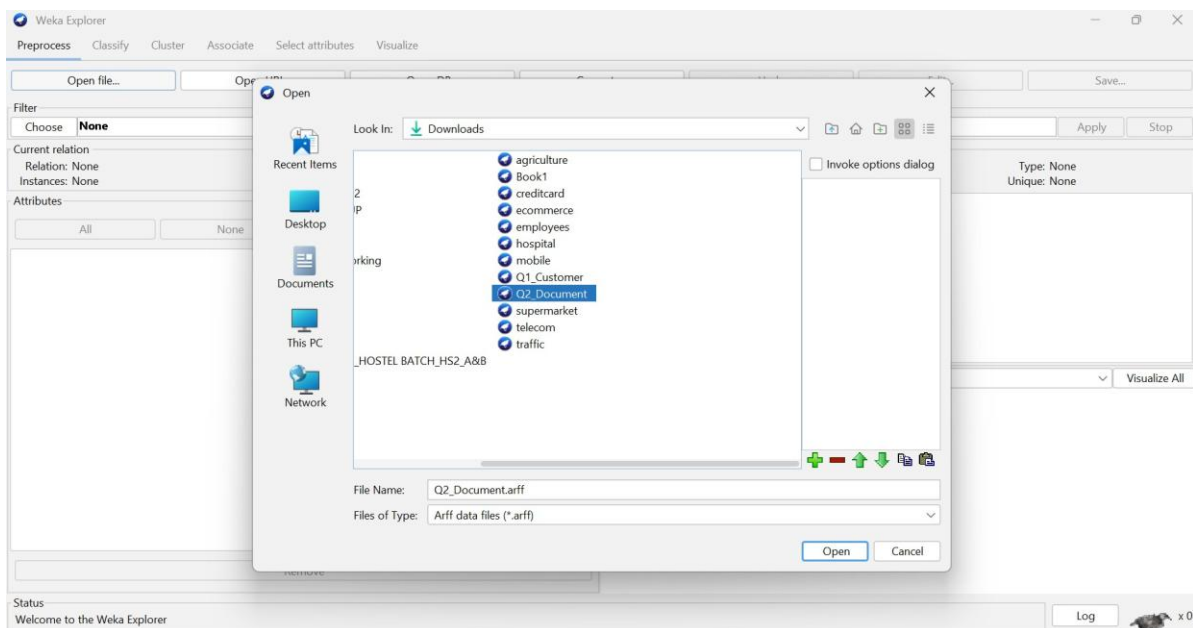
Compare K-means and EM algorithms in handling overlapping clusters for the given dataset.
(BL4 – 1 Mark)

Q₂ k-Means Vs. EM Algorithms

- * k-means assigns each document to only one cluster.
- * It works better when clusters are clearly separated.
- * EM gives a probability of a document belonging to different clusters.
- * Therefore, a document containing words from two topics can have membership in both clusters.

Conclusion:
EM is better than k-Means for handling overlapping document clusters.

OUTPUT:



Weka Explorer

Preprocess Classify **Cluster** Associate Select attributes Visualize

Clusterer Choose **SimpleKMeans** -init 0 -max-candidates 100 -periodic-pruning 10000 -min-density 2.0 -t1 -1.25 -t2 -1.0 -N 2 -A "weka.core.EuclideanDistance -R first-last" -I 500 -num-slots 1 -S 10

Cluster mode

- ☒ Use training set
- ☐ Supplied test set Set...
- ☐ Percentage split % 66
- ☐ Classes to clusters evaluation (Num) Word3
- ☒ Store clusters for visualization

Ignore attributes

Start Stop

Result list (right-click for options)

06:43:03 - SimpleKMeans

Clusterer output

Cluster 0: D4,0.75,0.15,0.1
Cluster 1: D2,0.1,0.6,0.3

Missing values globally replaced with mean/mode

Final cluster centroids:

Attribute	Full Data (5.0)	Cluster# 0 (2.0)	Cluster# 1 (3.0)
Document			
Word1	0.41	0.775	0.1667
Word2	0.35	0.125	0.5
Word3	0.24	0.1	0.3333

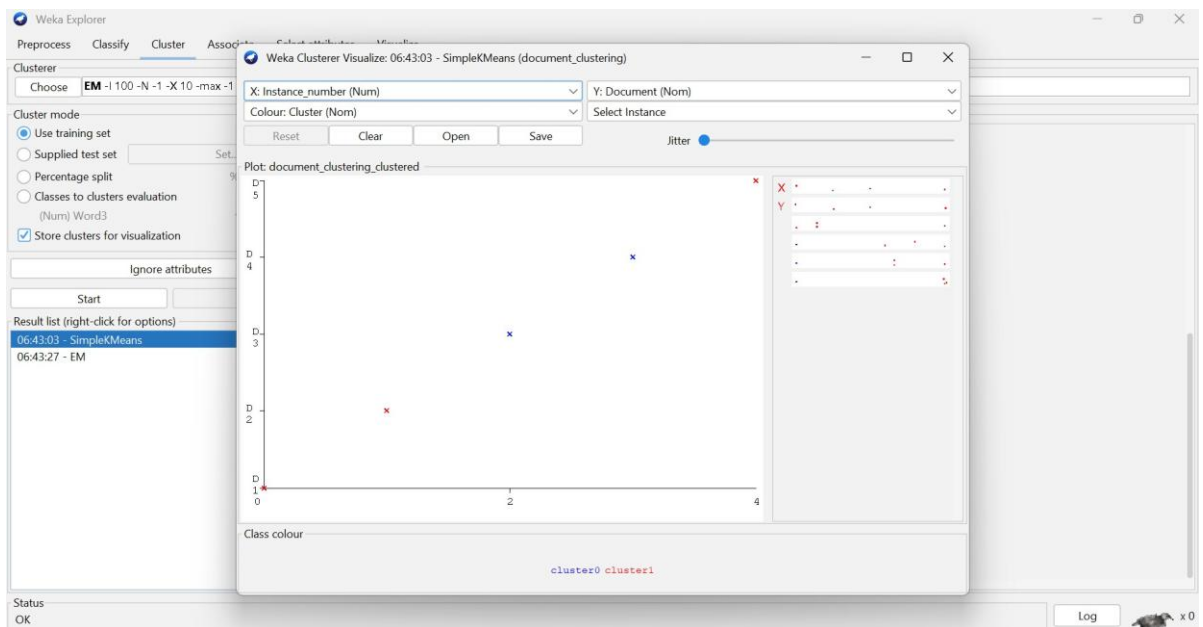
Time taken to build model (full training data) : 0.01 seconds

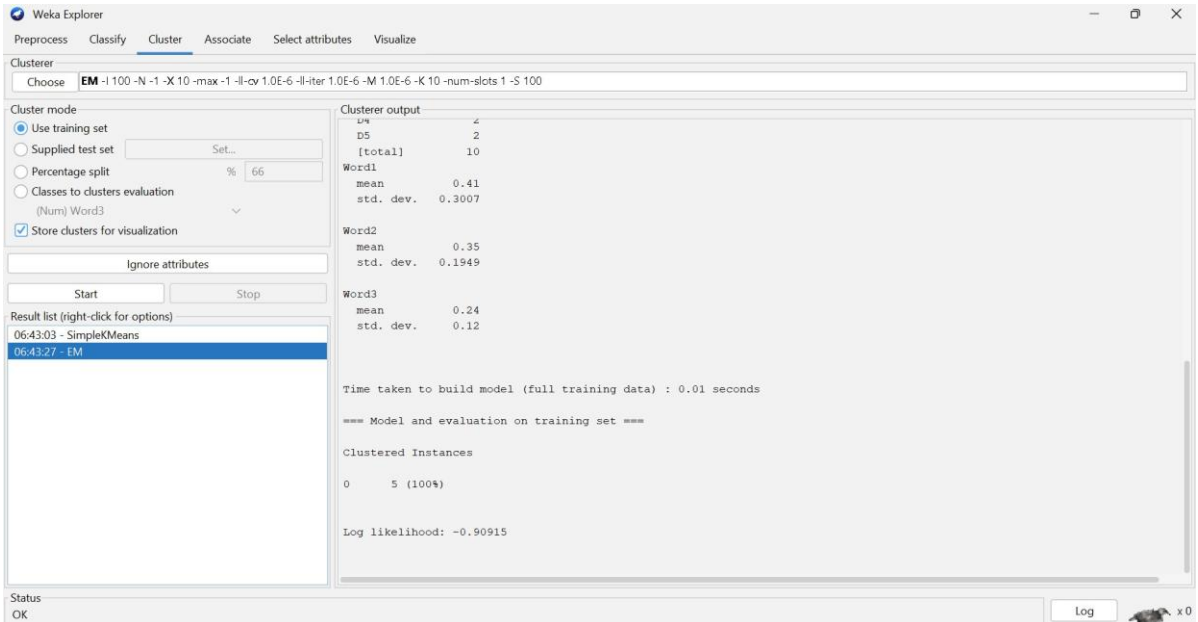
=== Model and evaluation on training set ===

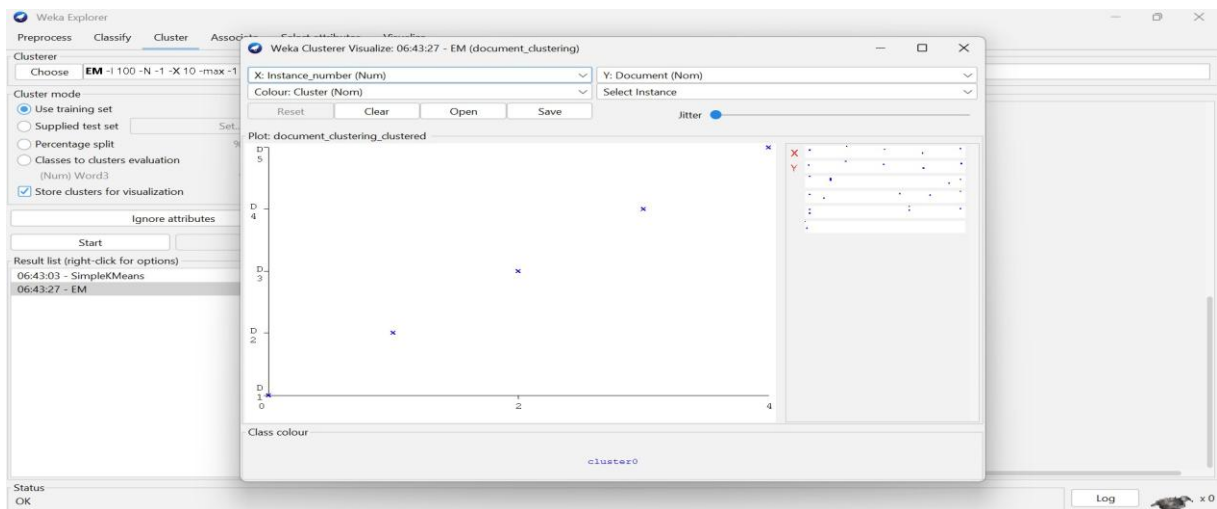
Clustered Instances

Cluster	Count	Percentage
0	2	40%
1	3	60%

Status OK Log x 0







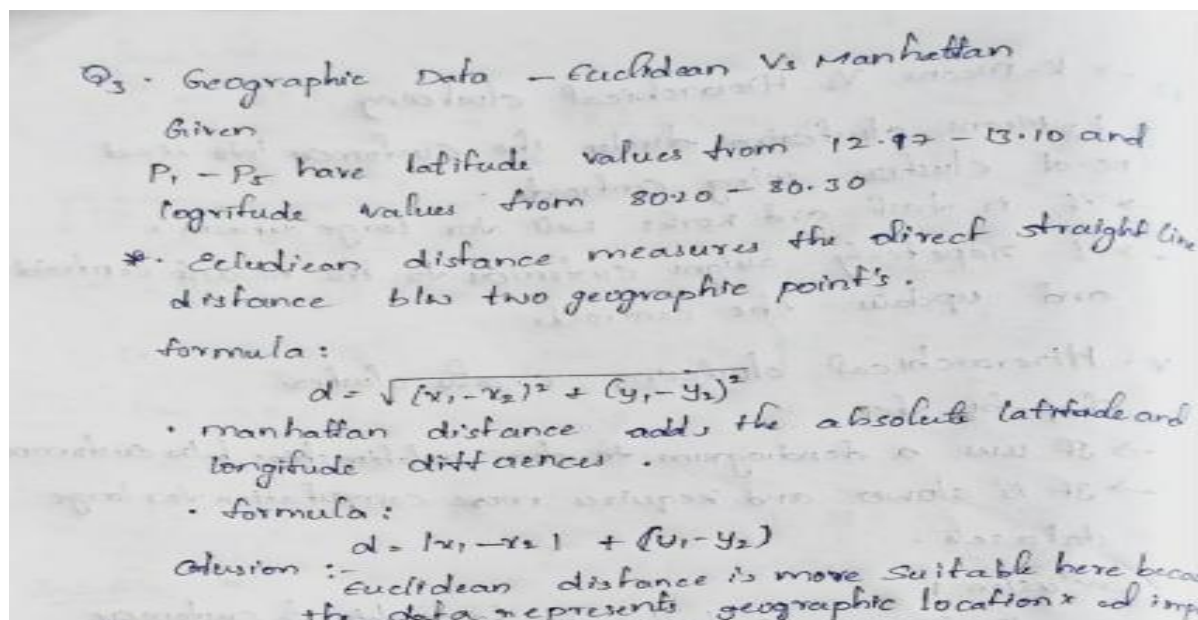
Q3. Dataset: Geographic Data

Point Latitude Longitude

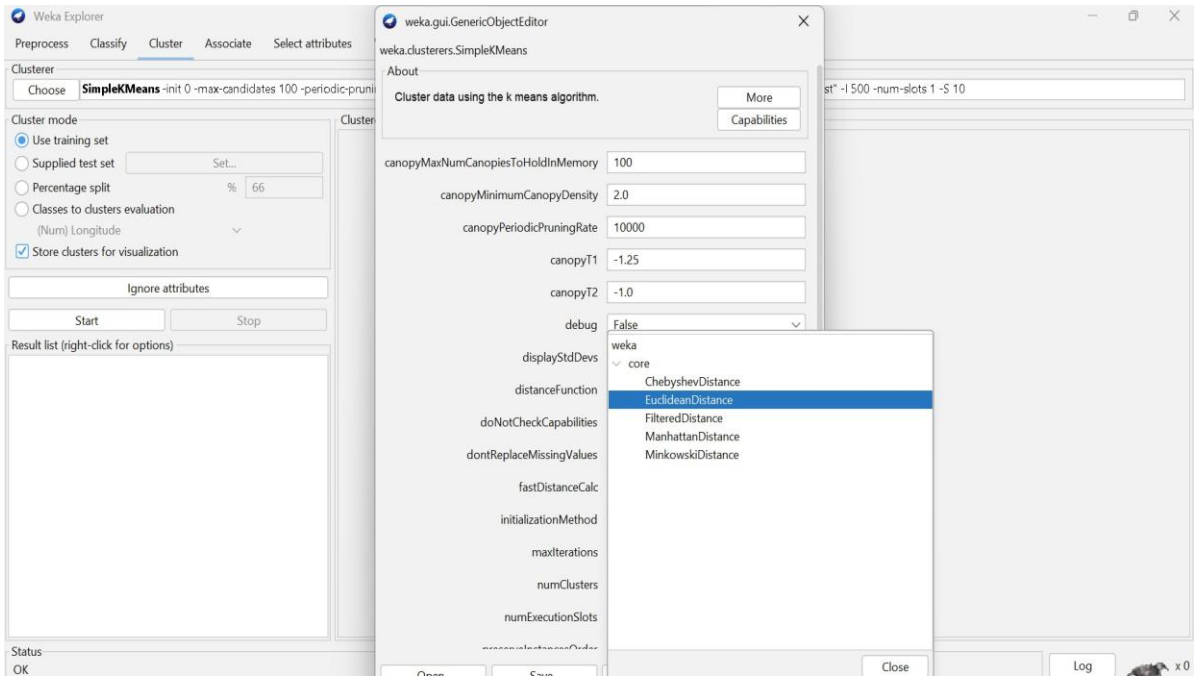
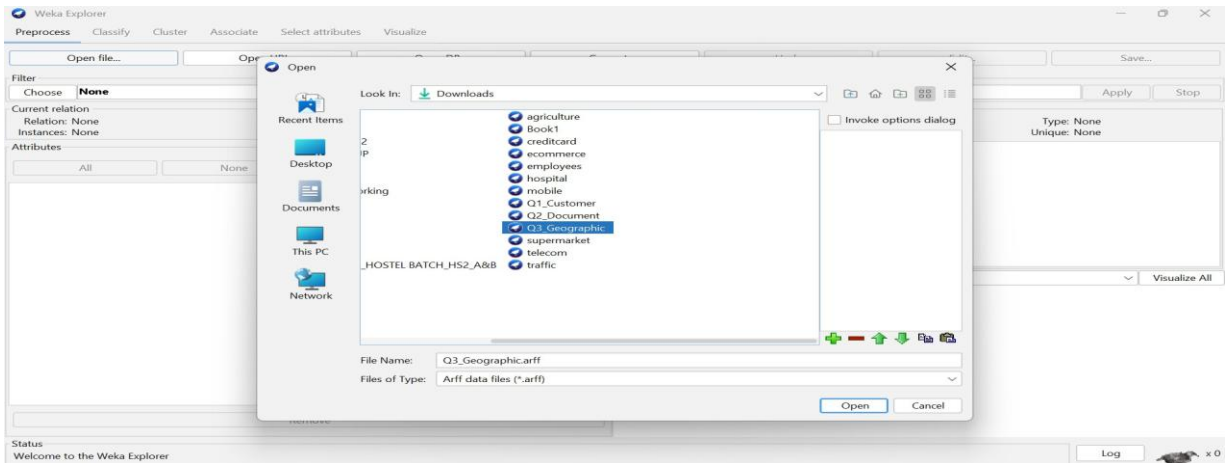
P1	13.08	80.27
P2	13.10	80.25
P3	12.97	80.22
P4	13.05	80.30
P5	13.00	80.20

Question:

Compare Euclidean and Manhattan distance metrics and analyze their impact on clustering results. (BL4 – 1 Mark)



OUTPUT:



Weka Explorer

Preprocess Classify **Cluster** Associate Select attributes Visualize

Cluster: Choose **SimpleKMeans** -init 0 -max-candidates 100 -periodic-pruning 10000 -min-density 2.0 -t1 -1.25 -t2 -1.0 -N 2 -A "weka.core.EuclideanDistance -R first-last" -I 500 -num-slots 1 -S 10

Cluster mode

- ☒ Use training set
- ☐ Supplied test set
- ☐ Percentage split
- ☐ Classes to clusters evaluation
- ☒ Store clusters for visualization

Ignore attributes

Start Stop

Result list (right-click for options)

06:48:47 - SimpleKMeans

Cluster output

Initial starting points (random):

Cluster 0: P4,13.05,80.3
Cluster 1: P2,13.1,80.25

Missing values globally replaced with mean/mode

Final cluster centroids:

Attribute	Full Data	Cluster#	0	1
	(5.0)		(2.0)	(3.0)
Point	P1	P3	P1	
Latitude	13.04	12.985	13.0767	
Longitude	80.248	80.21	80.2733	

Time taken to build model (full training data) : 0 seconds

=== Model and evaluation on training set ===

Clustered Instances

0	2 (40%)
1	3 (60%)

Status OK Log

Weka Explorer

Preprocess Classify **Cluster** Associate Select attributes Visualize

Cluster: Choose **SimpleKMeans** -init 0 -max-candidates 100 -periodic-pruning 10000 -min-density 2.0 -t1 -1.25 -t2 -1.0 -N 2 -A "weka.core.EuclideanDistance -R first-last" -I 500 -num-slots 1 -S 10

Cluster mode

- ☒ Use training set
- ☐ Supplied test set
- ☐ Percentage split
- ☐ Classes to clusters evaluation
- ☒ Store clusters for visualization

Ignore attributes

Start Stop

Result list (right-click for options)

06:48:47 - SimpleKMeans

weka.gui.GenericObjectEditor

About

Cluster data using the k means algorithm.

More Capabilities

canopyMaxNumCanopiesToHoldInMemory 100

canopyMinimumCanopyDensity 2.0

canopyPeriodicPruningRate 10000

canopyT1 -1.25

canopyT2 -1.0

debug False

displayStdDevs weka

distanceFunction core

doNotCheckCapabilities

dontReplaceMissingValues

fastDistanceCalc

initializationMethod

maxIterations

numClusters

numExecutionSlots

preserveInstancesOrder

Close Log

Weka Explorer

Preprocess Classify **Cluster** Associate Select attributes Visualize

Cluster: Choose **SimpleKMeans** -init 0 -max-candidates 100 -periodic-pruning 10000 -min-density 2.0 -t1 -1.25 -t2 -1.0 -N 2 -A "weka.core.ManhattanDistance -R first-last" -I 500 -num-slots 1 -S 10

Cluster mode

- ☒ Use training set
- ☐ Supplied test set
- ☐ Percentage split
- ☐ Classes to clusters evaluation
- ☒ Store clusters for visualization

Ignore attributes

Start Stop

Result list (right-click for options)

06:48:47 - SimpleKMeans

06:49:48 - SimpleKMeans

Cluster output

Initial starting points (random):

Cluster 0: P4,13.05,80.3
Cluster 1: P2,13.1,80.25

Missing values globally replaced with mean/mode

Final cluster centroids:

Attribute	Full Data	Cluster#	0	1
	(5.0)		(1.0)	(4.0)
Point	P1	P4	P1	
Latitude	13.05	13.05	13.04	
Longitude	80.25	80.3	80.235	

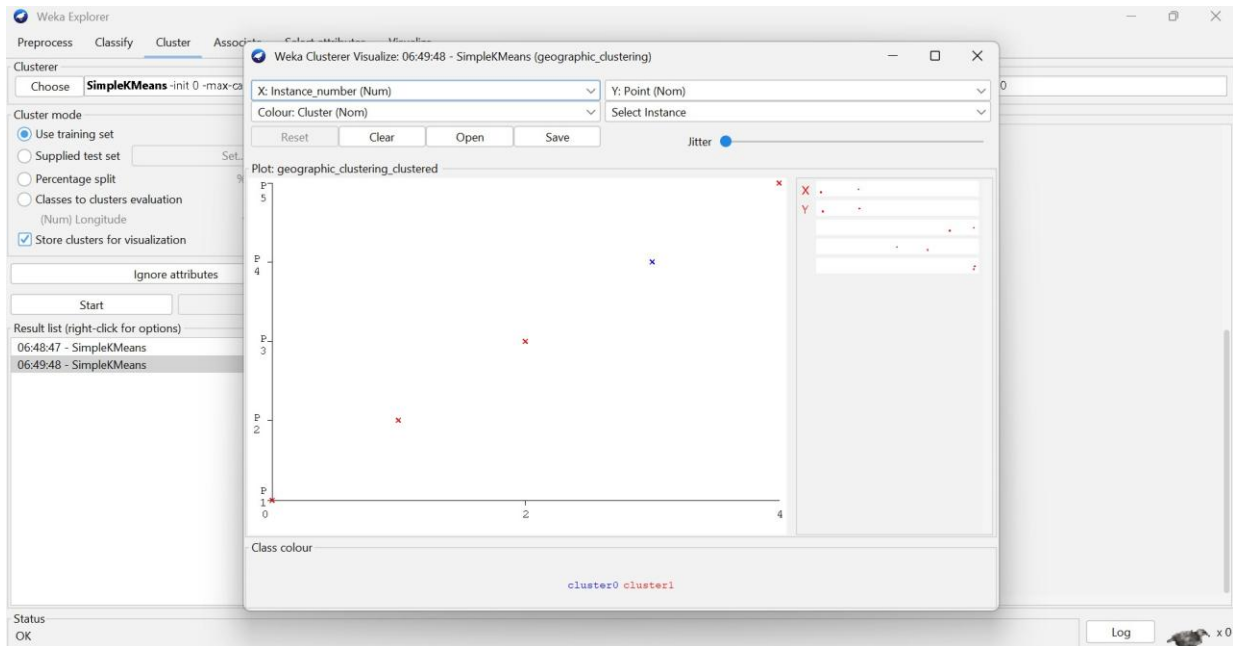
Time taken to build model (full training data) : 0 seconds

=== Model and evaluation on training set ===

Clustered Instances

0	1 (20%)
1	4 (80%)

Status OK Log



Q4. Dataset: Retail Products

Product Sales (₹) Frequency

P1	50000	20
P2	30000	10
P3	52000	22
P4	25000	8
P5	32000	12

Question:

Compare agglomerative and divisive clustering approaches and explain their suitability for this dataset. (BL3 – 1 Mark)

Q4.

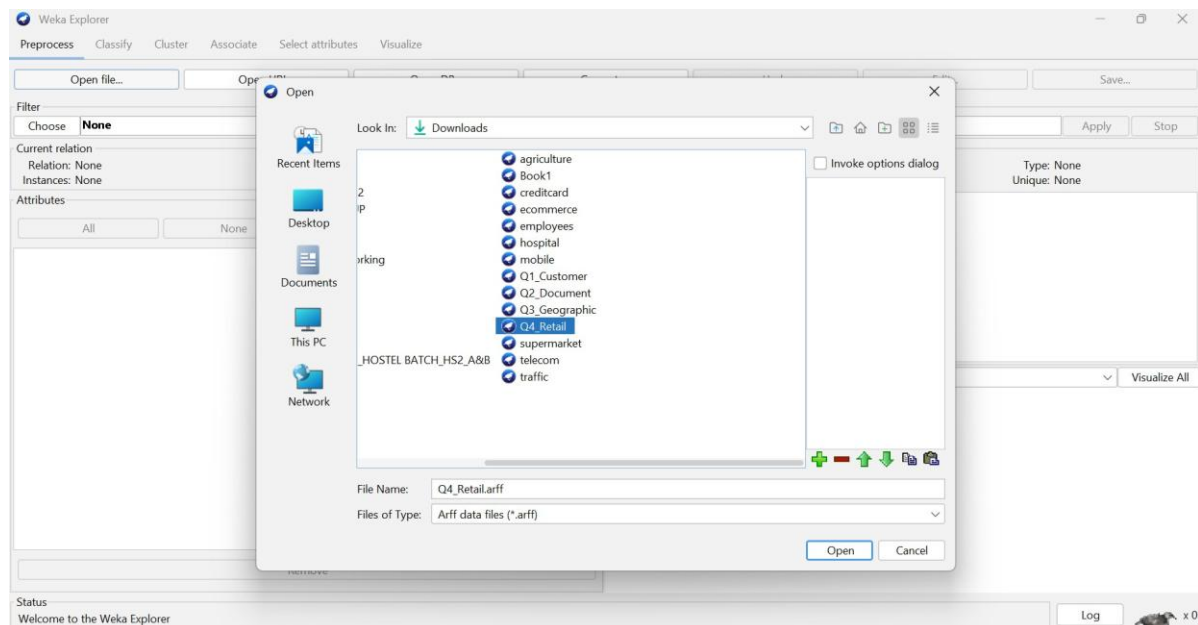
Retail products = Agglomerative vs Divisive

Given Sales = ₹20,000 - ₹52,000 and frequency

⇒ P22

- Agglomerative clustering starts with each product as separate cluster.
- It gradually merges similar products based on sales and frequency.
- Divisive clustering starts with all five products in cluster.
- It then splits the products into smaller groups.
- Agglomerative clustering is simple and suitable for grouping these retail products based on sales and purchase frequency.

OUTPUT:



Weka Explorer

Preprocess Classify **Cluster** Associate Select attributes Visualize

Clusterer
Choose **HierarchicalClusterer** -N 2 -L SINGLE -P -A "weka.core.EuclideanDistance -R first-last"

Cluster mode
☒ Use training set
☐ Supplied test set Set...
☐ Percentage split % 66
☐ Classes to clusters evaluation (Num) Frequency
☒ Store clusters for visualization

Ignore attributes
Start Stop

Result list (right-click for options)
 06:53:39 - HierarchicalClusterer

Clusterer output

```

Relation: retail_products
Instances: 5
Attributes: 3
Product
Sales
Frequency

Test mode: evaluate on training data

=== Clustering model (full training set) ===

Cluster 0
(20.0:1.01286,22.0:1.01286)

Cluster 1
((10.0:1.01286,12.0:1.01286):0.01412,8.0:1.02699)

Time taken to build model (full training data) : 0 seconds

=== Model and evaluation on training set ===

Clustered Instances

0 2 ( 40%)
1 3 ( 60%)
  
```

Status OK Log x0

Weka Explorer

Preprocess Classify **Cluster** Associate Select attributes Visualize

Clusterer
Choose **MakeDensityBasedClusterer** -M 1.0E-6 -W weka.clusterers.SimpleKMeans -- -init 0 -max-candidates 100 -periodic-pruning 10000 -min-density 2.0 -t1 -1.25 -t2 -1.0 -N 2 -A "weka.core.EuclideanDistance -R first-last" -I 500 -r

Cluster mode
☒ Use training set
☐ Supplied test set Set...
☐ Percentage split % 66
☐ Classes to clusters evaluation (Num) Frequency
☒ Store clusters for visualization

Ignore attributes
Start Stop

Result list (right-click for options)
 06:53:39 - HierarchicalClusterer
 06:54:50 - MakeDensityBasedClusterer

Clusterer output

```

Discrete Estimator. Counts = 2 2 1 2 2 (Total = 8)
Attribute: Sales
Normal Distribution. Mean = 29000 StdDev = 2943.9203
Attribute: Frequency
Normal Distribution. Mean = 10 StdDev = 1.633

Cluster: 1 Prior probability: 0.4286

Attribute: Product
Discrete Estimator. Counts = 2 1 2 1 1 (Total = 7)
Attribute: Sales
Normal Distribution. Mean = 51000 StdDev = 1000
Attribute: Frequency
Normal Distribution. Mean = 21 StdDev = 1

Time taken to build model (full training data) : 0 seconds

=== Model and evaluation on training set ===

Clustered Instances

0 3 ( 60%)
1 2 ( 40%)

Log likelihood: -12.6953
  
```

Status OK Log x0

Weka Explorer

Preprocess Classify **Cluster** Associate Select attributes Visualize

Clusterer
Choose **MakeDensityBasedClusterer**

Cluster mode
☒ Use training set
☐ Supplied test set Set...
☐ Percentage split % 66
☐ Classes to clusters evaluation (Num) Frequency
☒ Store clusters for visualization

Ignore attributes
Start Stop

Result list (right-click for options)
 06:53:39 - HierarchicalClusterer
 06:54:50 - MakeDensityBasedClusterer

Weka Clusterer Visualize: 06:54:50 - MakeDensityBasedClusterer (retail_products)

X: Instance_number (Num) Y: Product (Nom)
 Colour: Cluster (Nom) Select Instance
 Reset Clear Open Save Jitter

Plot: retail_products_clustered

Class colour cluster0 cluster1

Status OK Log x0

Q5 . Dataset: Mixed Attributes

Item	Price (₹)	Category	Rating
I1	500	A	4.5
I2	300	B	3.8
I3	550	A	4.7
I4	250	B	3.5
I5	520	A	4.6

Question:

Compare centroid-based and density-based clustering techniques and analyze their effectiveness on mixed data types.

Q5.

Mixed - Attributes - Centroid - Based Vs Density Based

Given :

price $\Rightarrow$ ₹250 - ₹550,

Category = A/B,

Rating = 3.5 - 4.7

→ The dataset contains numerical attributes such as price and Rating.

→ It also contains a categorical attribute, category (A/B)

→ Centroid-based clustering, such as k-means forms groups around a central point.

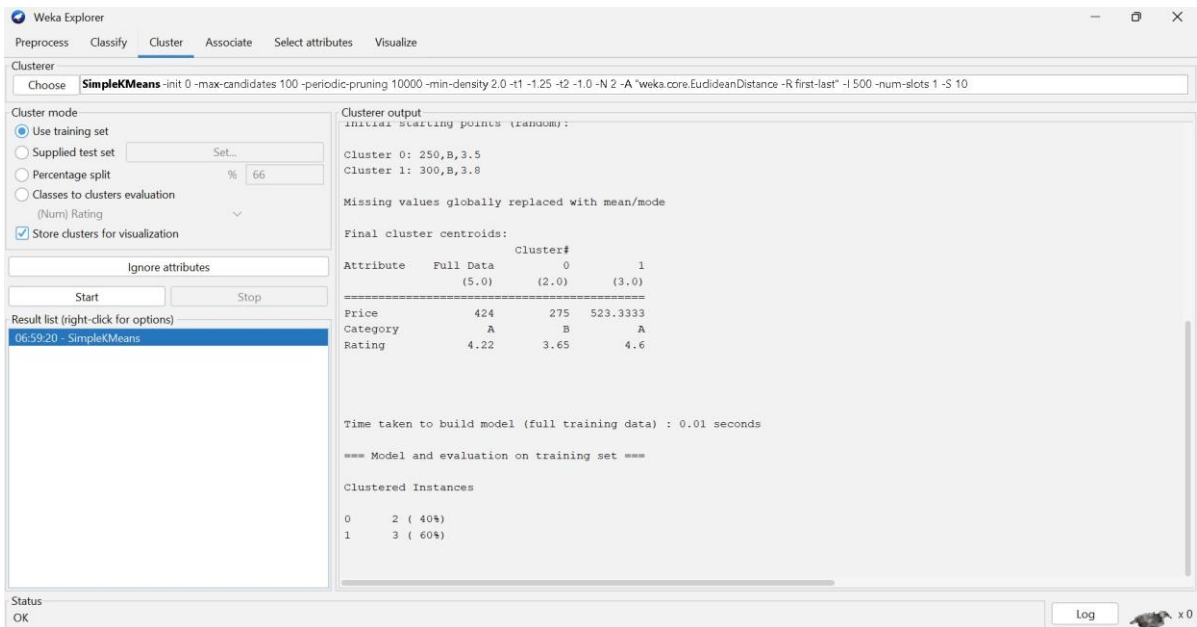
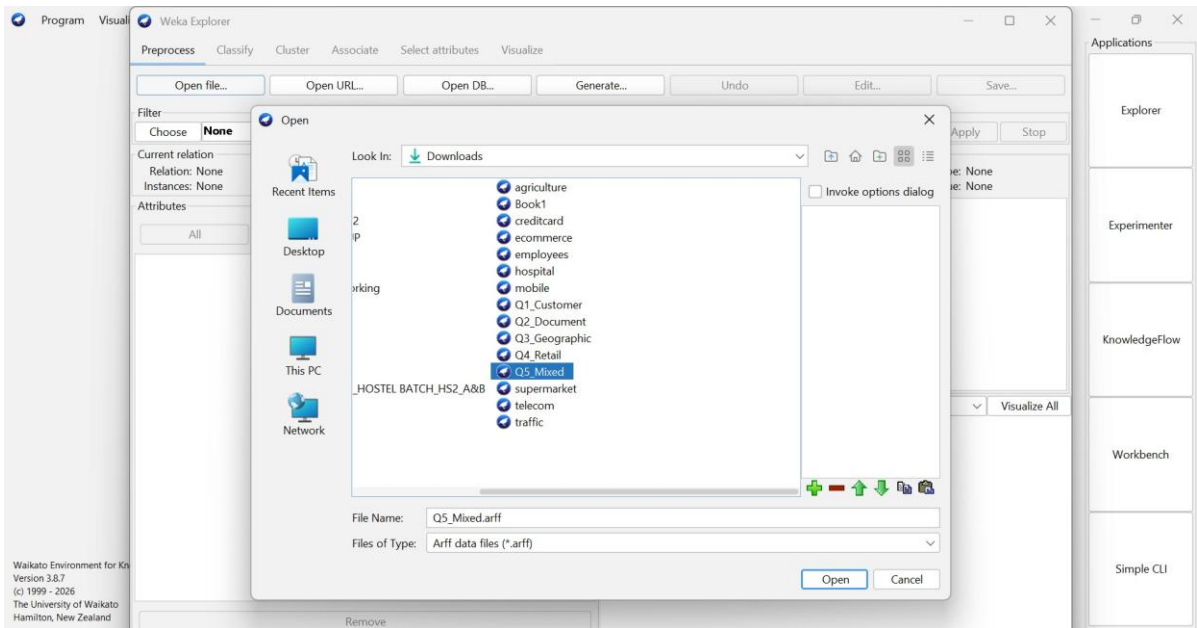
* k-means mainly works with numerical values, so Category must be properly encoded.

• It can also identify unusual product as outliers.

Conclusion:

Centroid-based clustering is simple for numerical attribute, while density-based clustering is useful when the product forms irregular group.

OUTPUT:



Weka Explorer

Preprocess Classify **Cluster** Associate Select attributes Visualize

Clusterer: Choose **MakeDensityBasedClusterer** -M 1.0E-6 -W weka.clusterers.SimpleKMeans -- -init 0 -max-candidates 100 -periodic-pruning 10000 -min-density 2.0 -t1 -1.25 -t2 -1.0 -N 2 -A "weka.core.EuclideanDistance -R first-last" -I 500 -n

Cluster mode

- ☒ Use training set
- ☐ Supplied test set Set...
- ☐ Percentage split % 66
- ☐ Classes to clusters evaluation (Num) Rating
- ☒ Store clusters for visualization

Ignore attributes

Start Stop

Result list (right-click for options)

- 06:59:20 - SimpleKMeans
- 06:59:39 - MakeDensityBasedClusterer**

Clusterer output

```
Normal Distribution. mean = 2.75 StdDev = 2.0
Attribute: Category
Discrete Estimator. Counts = 1 3 (Total = 4)
Attribute: Rating
Normal Distribution. Mean = 3.65 StdDev = 0.15

Cluster: 1 Prior probability: 0.5714

Attribute: Price
Normal Distribution. Mean = 523.3333 StdDev = 20.548
Attribute: Category
Discrete Estimator. Counts = 4 1 (Total = 5)
Attribute: Rating
Normal Distribution. Mean = 4.6 StdDev = 0.0816

Time taken to build model (full training data) : 0 seconds

=== Model and evaluation on training set ===

Clustered Instances

0      2 ( 40%)
1      3 ( 60%)

Log likelihood: -4.6007
```

Status OK Log x 0

